
Predictive and Generative Modeling of Therapeutic Antibody Developability Using Fine-Tuned Protein Language Models

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Abstract

Therapeutic antibody development is a costly and time-intensive process, with many candidates failing due to poor developability [1]. Developability metrics encompass biophysical properties such as thermostability, self-association propensity, hydrophobicity, and solubility [1]. These properties determine whether a candidate can be successfully manufactured and remain stable in clinical settings. In this project, we investigate whether antibody protein language models can (A) predict developability metrics and therapeutic approval from sequence alone, and (B) enable optimization of developability properties, while accounting for the binding affinity of the target antigen, through guided HCDR3 sequence generation.

For Task A, we evaluate frozen AbLang2 [2] embeddings with Ridge regression against LoRA-fine-tuned [3] AbLang2 with learned regression heads on the Ginkgo Bioworks GDPa1 dataset under hierarchical-cluster, IgG isotype-stratified 5-fold cross-validation [4]. Frozen AbLang2 with Ridge outperforms LoRA fine-tuning on the majority of 11 biophysical targets (mean $\Delta\rho = +0.067$ over the p-IgGen baseline), with the strongest correlations on HAC ($\rho = 0.703$), AC-SINS_pH7.4 ($\rho = 0.523$), and PR_Ova ($\rho = 0.514$). Both approaches yield only moderate approval classification performance (AUPRC ≤ 0.62) compared with the baseline XGBoost model using experimental features (AUPRC of 0.665).

For Task B, we design a three-phase generative pipeline: baseline AbLang2 only generation, flow-matched steering of AbLang2 for targets with better developability scores, and steering of AbLang2 for developability and binding affinity. Generated HCDR3 variants show statistically significant improvements over parent antibodies in oracle-predicted HIC and AC-SINS, CamSol solubility, and SAP score. Pretraining the flow model on a larger dataset (OAS) further improved the developability scores. The binding affinity prediction layer performed poorly; therefore, the contribution of binding guidance in the guided condition remains unquantified.

1 Introduction

Therapeutic antibodies are among the most widely used classes of protein-based biotherapeutics [2, 5]. Despite this commercial success, the development of therapeutic antibodies remains significantly time-consuming [5]. A central bottleneck is lead optimization, where a single candidate must be selected from a panel of molecules with comparable target binding affinity based on predicted manufacturability, stability, immunogenicity, and other biophysical properties (collectively termed

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developability) [1]. Poor developability properties can lead to late-stage clinical failure. This makes early identification of developability issues a priority for antibody manufacturing [6].

Biophysical properties that govern antibody developability include thermostability, self-association propensity, hydrophobicity, polyreactivity to off-target proteins, charge-mediated interactions, monomer purity, and expression titer [1, 4]. These properties are inherently encoded in the antibody’s sequence, particularly in the complementarity-determining regions (CDRs) that mediate antigen binding. This creates a fundamental tension between affinity optimization and developability [6].

Recent advances in protein language models (PLMs) have demonstrated that self-supervised pretraining on large protein sequence corpora can learn representations that encode structural and functional information [2]. Antibody-specific PLMs, such as AbLang2 [2], provide high-dimensional embeddings that capture antibody-specific structural context. Furthermore, while PLMs excel at prediction tasks, a complementary application is generative optimization using learned representations to guide the design of novel antibody sequences with improved developability profiles while preserving target binding properties.

Our central research question is: Can sequence-based antibody protein language models (A) predict developability metrics and downstream therapeutic approval from sequence alone, and (B) enable optimization of true developability properties through guided sequence generation?

To address these questions, we conduct a systematic investigation spanning multiple modeling approaches and evaluation frameworks. Our main contributions are below:

- We formulate both developability regression and therapeutic approval classification as sequence-only prediction tasks using the Ginkgo Bioworks GDPa1 dataset, and evaluate frozen AbLang2 embeddings with Ridge regression against LoRA-fine-tuned AbLang2 with learned regression heads under hierarchical-cluster, IgG isotype-stratified 5-fold cross-validation.
- We implement a three-phase generative pipeline for HCDR3 developability optimization using flow-matched steering of AbLang2 for targets with better developability scores and steering of AbLang2 accounting for developability and binding affinity. Then, we compare this approach against an unguided AbLang2 baseline and parent sequences across multiple biophysical evaluation metrics.
- We demonstrate improvements in oracle-predicted developability scores (HIC, AC-SINS) and independent sequence-based solubility and aggregation metrics (CamSol, SAP) for generated sequences, and analyze failure modes through systematic ablation studies (frozen vs. fine-tuned embeddings, CDR3-only vs. full-sequence input, OAS pretraining of flow model) and error analysis linking regression difficulty to target coefficient of variation.

2 Background

2.1 Antibody Developability and the Need for Sequence-only Prediction

Developability refers to the biophysical properties an antibody candidate must satisfy to be successfully manufactured and safely administered [1]. The most clinically relevant metrics include thermal stability (T_{m1} , T_{m2}), hydrophobicity and non-specific surface binding (HIC, SMAC, HAC, SAP score), self-association and polyreactivity (AC-SINS, PR_CHO, PR_Ova), and colloidal stability (SEC %Monomer, Expression Titer) [1, 4]. A key limitation of prior predictive work (Arsiwala et al. [4]) is its reliance on experimentally measured biophysical features as inputs. Since these features are available only after experimental characterization, predicting approval from sequence alone is more clinically relevant for early-stage lead evaluation.

2.2 Protein Language Models and Antibody Language Models

Antibody language models pretrained on large sequence corpora can encode structural and biophysical information without explicit physics-based calculations [7]. Linear probing of pretrained embeddings with Ridge regression is a strong and interpretable baseline for developability prediction, with

Spearman correlations of 0.3-0.7 reported for thermostability on small datasets [8]. However, general-purpose protein language models such as ESM2 exhibit a well-documented germline bias. Due to pretraining bias toward germline-proximal sequences, ESM2 assigns low likelihood to somatic hypermutations in CDR regions, which are essential for antigen specificity [9]. AbLang2 processes paired heavy- and light-chain sequences using a bidirectional transformer architecture and was designed to correct for germline bias by downweighting germline sequences during pretraining. This allows the model to produce embeddings that represent the somatic diversity present in therapeutic antibodies [9].

2.3 Conditional Flow Matching

Conditional flow matching (CFM) [10] has recently emerged as a framework for generative modeling, learning continuous-time velocity fields that transport prior distributions toward target distributions. Unlike diffusion models (which require many denoising steps), flow matching models generate samples through efficient ODE integration in a learned latent space [10]. This makes CFM attractive for protein design tasks where the target distribution is defined by a small number of experimentally characterized sequences, and where conditioning on developability properties can be incorporated into the velocity field.

2.4 Generative Models for Antibody CDR Optimization

A recent approach from Ginkgo Bioworks trains a discrete diffusion model on 1.68 million paired OAS sequences to generate developable CDR3 variants [11]. This pipeline inherits ESM2’s germline bias, potentially limiting its ability to model the somatic mutations most relevant to developability optimization. A related approach, ProtFlow [12], trains a continuous flow matching model in a compressed protein language model embedding space for fast protein sequence design. We build on both of these ideas: we use AbLang2 as the backbone (providing embeddings better suited to the therapeutic antibody distribution) and replace discrete diffusion with continuous conditional flow matching [12]. Flow matching trains a velocity network to transport Gaussian noise to the data distribution along straight-line interpolant trajectories, rather than the curved, noise-schedule-dependent paths of score-based diffusion, yielding faster training convergence and more stable ODE integration at inference. FiLM (Feature-wise Linear Modulation) [13] conditioning is used to steer generation toward desired developability targets. Prior models focused on improving developability did not account for maintaining or improving binding affinity to the target antigen. To address this gap, we built an architecture to jointly optimize antibody developability while preserving or improving target binding affinity in the HCDR3 region.

3 Methods

3.1 Ginkgo Bioworks PROPHET-Ab GDPa1

Our primary dataset is the Ginkgo Bioworks PROPHET-Ab GDPa1 [4], a publicly available high-throughput platform dataset comprising 246 therapeutic antibodies with experimentally measured biophysical properties and paired variable heavy (VH) and variable light (VL) chain amino acid sequences. The antibodies included in GDPa1 comprise 106 FDA-approved, 135 clinical-stage, and 5 withdrawn therapeutic antibodies. These antibodies primarily span IgG1, IgG2, and IgG4 isotypes. Experimentally derived developability metrics include HIC (Hydrophobic Interaction Chromatography) retention time, SEC %Monomer, SMAC (surface membrane affinity chromatography), HAC (hydrophobic affinity chromatography), polyreactivity against CHO host cell proteins (PR_CHO) and ovalbumin (PR_Ova), AC-SINS at pH 6.0 and 7.4, melting temperatures T_{m1} and T_{m2} , and expression Titer.

Developability Metric and Approval Prediction Preprocessing: We filtered the GDPa1 dataset down to 198 antibodies with binary approval labels (106 approved, 92 not approved). Missing biophysical values were imputed with the per-feature median. HAC showed the highest missingness (129/198, ~65%), followed by PR_CHO, PR_Ova, and T_{m2} ($n = 41$ each, ~21%); T_{m1} was missing for 9 antibodies, and the remaining six features (HIC, SMAC, SEC %Monomer, AC-SINS pH 6.0, AC-SINS pH 7.4, Titer) were each missing for fewer than 5.

Generative Modeling Preprocessing: For the generative pipeline, fold 1 (50 antibodies) was held out for testing, and folds 2–5 (196 antibodies) were used for training, following the GDPa1 dataset’s provided hierarchical-cluster, IgG isotype-stratified 5-fold splits. Sequences for the VH domain (average 110-130 residues) and VL domain (average 100-120 residues) were available for all 246 antibodies. HCDR3 lengths range from 5 to 24 residues (median 13), with no significant distribution shift between the training and test folds. The missing values in HIC and AC-SINS were imputed with the training median. Three types of embedding were computed per antibody: HCDR3 region, framework region (excluding HCDR3 region), and full sequence.

AbRank Affinity Dataset Preprocessing: To train the antigen-binding affinity component of the generative pipeline, we used the AbRank dataset [14], a curated collection of antibody–antigen pairs with experimentally derived binding affinities expressed as log-transformed affinity labels ($\log_{10} K_d$). After removing entries with missing heavy-chain sequence, light-chain sequence, antigen sequence, or affinity label, the dataset yielded about 169,000 unique antibody–antigen pairs. HCDR3 sequences were extracted from AbRank heavy chains using ANARCI AHo alignment, applying the same extraction logic used for the GDPa1 sequences to ensure consistency. Two training subsets were derived: (i) *Stage 1 (pretraining)*: all rows with a valid `log_Aff`, deduplicated on the tuple (VH, VL, antigen sequence) by averaging affinities, yielding about 167,000 pairs used for mean-pool antigen MSE pretraining; and (ii) *Stage 2 (fine-tuning)*: a subset of about 8,000 pairs from groups sharing the same framework scaffold and antigen sequence but differing in HCDR3 identity (i.e., multi-variant groups), enabling the model to learn how CDR3 variation modulates binding affinity against a fixed antigen and framework (i.e., excluding HCDR3 region). Target antigen sequences for the 47 clinical mAbs with known binding partners were retrieved from UniProt and cached as ESM2-650M per-residue embeddings.

3.2 Approval Prediction Baseline: XGBoost on Experimental Features

An XGBoost binary classifier [15] was trained on HIC, SMAC, HAC, PR_Ova, PR_CHO, SEC %Monomer, AC-SINS_pH6.0, AC-SINS_pH7.4, T_{m1} , T_{m2} , and Expression Titer. The model was configured with `max_depth = 25`, `learning_rate = 0.25`, `n_estimators = 200`, `subsample = 0.8`, `colsample_bytree = 0.8`, `reg_alpha = 0.1`, `reg_lambda = 0.1`, and `objective = binary:logistic`. This configuration matches the XGBoost hyperparameters reported in Arsiwala et al. [4]. Features were standardized using StandardScaler fitted on the training split only. Missing feature values were imputed with the per-feature median.

For training and evaluation, 20 independent XGBoost models were trained using random stratified 80:20 train-test splits. Each model produced a predicted approval probability for its test antibodies. The per-antibody developability score is defined as the mean predicted probability averaged across all bootstrap runs in which that antibody appears in the test set. A null model was evaluated identically using permuted labels (permutation seed matched to train-test split seeds). The primary evaluation metrics were AUPRC and AUROC (AUPRC was favored due to class imbalance properties in the GDPa1 dataset). The separation between approved and non-approved developability scores was assessed using a one-sided Mann-Whitney U-test. Feature importance scores were computed by training a single model on the full dataset (using `random_state = 42`).

3.3 Developability Metric Prediction Baseline: Ridge Regression on p-IgGen Embeddings

For each developability metric, a Ridge model (scikit-learn default $\alpha = 1.0$) was trained on mean-pooled p-IgGen embeddings [16]. Embeddings were computed by concatenating the VH and VL protein sequences with special begin ("1") and end ("2") tokens. Spaces were inserted between amino acid characters in the sequence string, followed by tokenizing with p-IgGen. The default forward pass through the model in batches of 16 was run, followed by extracting the final hidden layer and mean-pooling across all token positions (including the padding tokens).

Per-target dropping of NA values was applied so that each metric utilizes a subset of antibodies with non-missing labels. Two evaluation protocols were used: (i) a single random 80:20 train-test split (`random_state=42`), and (ii) a leave-one-fold-out cross-validation using the hierarchical_cluster_IgG_isotype_stratified_fold column provided in GDPa1 (partitions antibodies into 5 folds balanced by sequence similarity and IgG isotype). In cross-validation, the Ridge model is retrained from scratch on the four training folds for each held-out test fold. The cross-validation predictions

are aggregated across all folds, and an overall Spearman ρ is computed between all concatenated true and predicted values.

3.4 LoRA Fine-Tuning of AbLang2

We fine-tuned the AbLang2-paired encoder with Low-Rank Adaptation (LoRA) [3], injecting rank-8 adapters ($\alpha = 16$, dropout 0.1) into the query, key, value, and output projections of each self-attention block. All non-LoRA encoder weights were frozen. Paired (VH, VL) sequences were tokenized with the AbLang2 tokenizer, and a 480-dimensional embedding was obtained by mean-pooling the final hidden states across non-padding positions. For approval classification, a three-layer MLP head ($480 \rightarrow 256 \rightarrow 64 \rightarrow 1$, GELU, dropout 0.2) was trained with BCE-with-logits loss using a positive class weight of $N_{\text{neg}}/N_{\text{pos}}$. For regression, a two-layer MLP head ($480 \rightarrow 128 \rightarrow 1$) was trained with Huber loss ($\delta = 1.0$) on z -scored targets (StandardScaler fit on the training fold only). A separate head and adapter set is trained per target.

Training was performed in two phases. In Phase 1 (5 epochs), LoRA gradients are disabled, and only the head is trained with the AdamW optimizer ($\text{lr} = 1 \times 10^{-3}$, weight decay $= 10^{-2}$). In Phase 2 (45 epochs), LoRA gradients are enabled, where the head and adapters are trained jointly with discriminative learning rates (5×10^{-5} for adapters, 1×10^{-3} for the head), 3-epoch linear warmup followed by cosine decay, batch size 8, and gradient clipping at max norm 1.0. For approval, 15% of each training fold is held out as a stratified validation split for early stopping on validation AUPRC (patience 10). Regression runs the full 50 epochs without early stopping. Evaluation uses the same IgG isotype-stratified 5-fold cross-validation as Section 3.3, with a new AbLang2-paired model loaded per fold. All runs used seed 42.

3.5 Generative Pipeline

Oracle and Latent Space: Two Ridge regression oracles predict HIC and AC-SINS from 480-dimensional AbLang2 seqcoding embeddings of the paired (VH, VL) sequence, standardized with a StandardScaler fit on the 196 training antibodies. Partial Least Squares (PLS) regression with 10 components is then fitted on the training-set CDR3 embeddings (480d) supervised by HIC and AC-SINS jointly, reducing the flow model’s working space to only 10 maximally covariant dimensions with developability. All scalers are fit on training folds only. Oracle generalization is verified by 4-fold cross-validation within training folds with separate StandardScaler refits per inner loop.

Flow Model: A FiLM-conditioned MLP velocity network is trained with a conditional flow matching (CFM) objective in the 10-dimensional PLS-reduced CDR3 embedding space. The network takes three inputs: (i) the current 10-dimensional PLS position target embedding; (ii) a 480-dimensional masked framework embedding (HCDR3 positions replaced with the AbLang2 mask token ‘*’), projected to FiLM scale-and-bias via a two-layer MLP; and (iii) a 2-dimensional scaled developability target (HIC and AC-SINS z -scores), projected to a 128-dimensional residual. Two hidden layers (width 128, GELU, LayerNorm, 20% dropout) produce the 10-dimensional output velocity. $5 \times$ Gaussian noise augmentation ($\sigma_{\text{PLS}}=0.05$, $\sigma_{\text{fw}}=0.01$) expands the training set to 1,176 examples. Training uses MSE loss on the CFM velocity target ($\sigma=0.5$), AdamW ($\text{lr} = 10^{-3}$, weight decay $= 10^{-4}$), cosine annealing over 1,000 epochs (minimum $\text{lr} = 10^{-5}$), batch size 32, and gradient clipping at max norm 1.0.

Binding Scorer: A cross-attention network predicts $-\log K_d$ (higher = tighter binding) from the 480d AbLang2 seqcoding embedding and the ESM2-650M per-residue antigen embedding (1,280D, padded to 512 positions). An antibody projection head (linear + LayerNorm, $480 \rightarrow 128$) and an antigen projection head (linear + LayerNorm, $1,280 \rightarrow 128$) feed into 4-head cross-attention; the query and attended context are concatenated and scored by a two-layer MLP ($256 \rightarrow 128 \rightarrow 1$, GELU, dropout 0.1). Training uses AbRank data in two stages: Stage 1 (100 epochs, batch 64, $\text{lr} = 2 \times 10^{-4}$) trains on Ab-Ag pairs using mean-pooled ESM2 antigen vectors; Stage 2 (50 epochs, batch 16, $\text{lr} = 5 \times 10^{-6}$) fine-tunes on multi-variant pairs (same framework and antigen, differing HCDR3) using full per-residue antigen tensors. Both stages use AdamW (weight decay $= 10^{-4}$), cosine annealing, gradient clipping, and a 90:10 train/validation split for Spearman ρ test.

Generation Pipeline: The pipeline operates in three phases for each test antibody.

(i) **Phase 1 — ODE target sampling.** $N=50$ PLS targets are produced by 100-step Euler integration of the flow model from Gaussian noise, conditioned on the masked framework embedding and the top-20th-percentile developability target. For the *guided* condition, the velocity interpolates between binding scores and developable scores, and for flow only condition, it only uses the developability score. The mean of the $N=50$ endpoints initializes the PLS centroid z_{PLS}^* for Phase 2.

For the *guided* condition, the Euler step augments the developability-directed flow velocity with a binding gradient term, jointly steering the target toward both high-developability and high-affinity regions of PLS space (see Appendix A for the full update equation). For the *flow-only* condition, only flow velocity is used, steering purely by developability. The mean of the $N=50$ endpoints initializes the PLS centroid z_{PLS}^* for Phase 2.

(ii) **Phase 2 — Iterative MLM-steered sampling.** The parent VH is masked at HCDR3 positions and AbLang2 MLM log-probabilities are computed as a position-independent base distribution. For each CDR3 position, the top- k ($k=5$) tokens are evaluated with:

$$\text{bonus}(a) = -\alpha \cdot \|\hat{z}_{\text{PLS}} - z_{\text{PLS}}^*\|_2 + \beta \cdot s_{\text{bind}}(a), \quad (1)$$

where $\hat{z}_{\text{PLS}}$ is the PLS projection of the seqcoding embedding of the trial sequence (amino acid a at position i , other CDR3 positions fixed at MLM argmax), s_{bind} is the binding score, and $\alpha=\beta=1.0$. In this context, α represents the weight given to the developability score, while β represents the weight given to the binding affinity. Top- k base probabilities are scaled by $1 + \exp(\gamma(\text{bonus}(a) - \max_{a'} \text{bonus}(a')))$, $\gamma=2.0$; non-top- k tokens retain their original weight. $M=20$ candidates are drawn per round at $T=1.0$ and scored by $z_{\text{joint}} = z_{\text{HIC}} + z_{\text{SINS}} - s_{\text{bind}}$. If the antigen is found not to bind using the binding prediction layer, the candidate is rejected. After each round, z_{PLS}^* is updated from the top-5 candidates. The *flow-only* ablation sets $\beta=0$ with no threshold.

(iii) **Final selection.** The sequence with the lowest z_{joint} across $R=5$ rounds is retained. Four conditions are compared by paired t -test on $n=49$ test antibodies with retrievable antigen sequences: *parent* (unmodified), *baseline* (AbLang2 only), *flow-only* ($\lambda=0$, $\beta=0$), and *guided: flow + binding* ($\lambda=0.5$, $\beta=1.0$). CamSol solubility and sequence SAP score serve as independent validation metrics not used during training or candidate selection.

4 Results

4.1 Approval and Developability Metric Baseline Results

Approval Prediction: Table 1 summarizes the bootstrapped XGBoost results. The model achieved a median AUPRC of 0.665 and median AUROC of 0.622 across 20 bootstrap runs, compared to the null model’s median AUPRC of 0.593 and AUROC of 0.524. The mean developability score was 0.619 ± 0.285 for approved antibodies versus 0.477 ± 0.316 for non-approved antibodies (one-sided Mann-Whitney $p = 7.3 \times 10^{-4}$), consistent with the separation reported by Arsiwala et al. (0.60 vs. 0.49, $p = 3.5 \times 10^{-5}$) [4]. Feature importance analysis identified HAC, T_{m2} , T_{m1} , and PR_Ova as the four most informative features (Table 2). Each of these features exceeded the mean feature importance of 0.091.

Table 1: XGBoost approval prediction: bootstrapped performance over 20 stratified 80:20 splits. AUPRC is the primary metric. The null positive rate is 0.535.

Model	Median AUPRC	Median AUROC
XGBoost (experimental features)	0.665	0.622
Null (shuffled labels)	0.593	0.524

Developability Metric Prediction: Ridge regression on p-IgGen embeddings was evaluated under both random 80:20 splits and 5-fold hierarchical-cluster cross-validation (Table 3). Random-split estimates were consistently inflated relative to cross-validation (e.g., HIC: 0.408 vs. 0.335; Titer: 0.477 vs. 0.239), confirming that sequence similarity inflates random-split metrics. Under cross-validation, the strongest p-IgGen signals were PR_Ova ($\rho = 0.549$), PR_CHO ($\rho = 0.496$), and AC-SINS_pH7.4 ($\rho = 0.394$). HAC reached $\rho = 0.621$ but with a smaller sample size ($n = 94$), while T_{m2} and SEC %Monomer were near zero or negative, indicating these properties are not reliably recoverable from p-IgGen embeddings alone. The full p-IgGen cross-validation values appear as the rightmost column of Table 5, serving as the baseline against the AbLang2 approaches.

Table 2: XGBoost feature importances (mean decrease in impurity), single model trained on all 198 labeled antibodies. Mean importance = 0.091.

Feature	Importance	Feature	Importance
HAC	0.166	Titer	0.072
T_{m2}	0.128	PR_CHO	0.070
T_{m1}	0.106	SMAC	0.068
PR_Ova	0.102	AC-SINS pH 7.4	0.066
SEC %Monomer	0.083	AC-SINS pH 6.0	0.063
HIC	0.075		

Table 3: Ridge regression on p-IgGen embeddings: Spearman ρ for 10 developability metrics under random 80:20 split and 5-fold hierarchical cluster, IgG isotype-stratified cross-validation. N = antibodies with non-missing labels. Cross-validation ρ is the primary reported figure; rows sorted by cross-validation ρ .

Target	N	Random-split ρ	Cross-Validation ρ
HAC	94	0.329	0.621
PR_Ova	197	0.598	0.549
PR_CHO	197	0.438	0.496
AC-SINS_pH7.4	242	0.456	0.394
HIC	242	0.408	0.335
Titer	239	0.477	0.239
SMAC	242	0.250	0.207
Purity	242	-0.019	0.180
T_{m2}	193	-0.111	-0.084
SEC %Monomer	242	0.174	-0.101

4.2 Approval and Developability Modeling

We used three approaches for antibody developability prediction: (1) AbLang2-LoRA (fine-tuned AbLang2 with LoRA adapters), (2) AbLang2 (frozen) with Ridge (frozen AbLang2 embeddings with Ridge regression), and (3) the Ridge with p-IgGen baseline. All models were evaluated using 5-fold cross-validation, with folds defined by hierarchical sequence clustering.

Approval Prediction: The best AUPRC was achieved by the Ridge baseline (0.621 ± 0.123), with AbLang2-LoRA performing slightly worse (0.574 ± 0.089). AUROC values were similarly modest for both models: (0.555 ± 0.070 for Ridge, 0.485 ± 0.076 for LoRA). These results indicate that, while sequence-level features contain some signal for approval, the task remains challenging and is not substantially improved by fine-tuning.

Table 4: Approval Prediction performance (5-fold cross-validation). AUPRC is the primary metric due to high class imbalance in GDPa1.

Model	AUPRC Mean	AUPRC Std. Dev	AUROC Mean	AUROC Std. Dev
AbLang2-LoRA (fine-tuned)	0.574	0.089	0.485	0.076
AbLang2 (frozen) + Ridge	0.621	0.123	0.555	0.070

Developability Metric Regression: Table 5 reports Spearman ρ for each developability target across AbLang2 with LoRA, AbLang2 (frozen) with Ridge, and p-IgGen with Ridge.

4.3 Generative Pipeline Results

Oracle performance: The Ridge regression oracle trained on AbLang2 full-sequence embeddings achieves Spearman $\rho = 0.32$ for HIC and $\rho = 0.445$ for AC-SINS on the held-out test set from GDPa1. The oracle is trained on full-sequence embeddings because HIC and AC-SINS are whole-molecule properties.

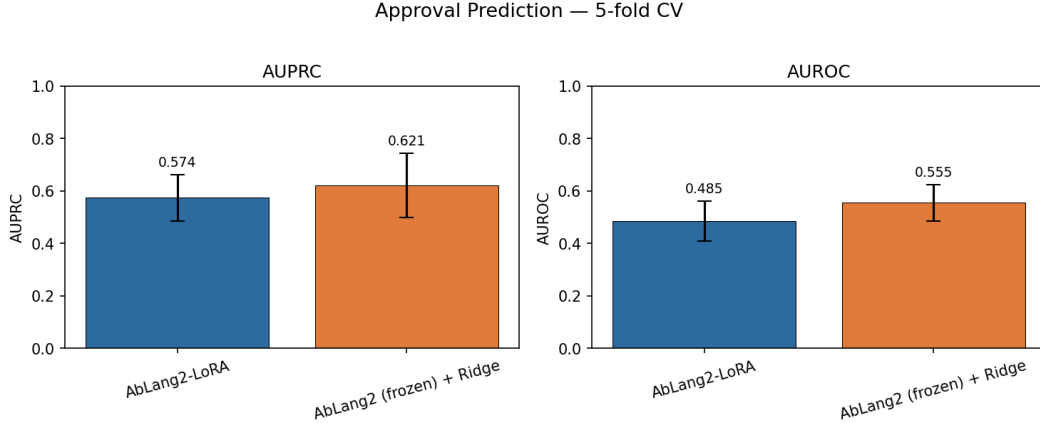


Figure 1: Mean and standard deviation of AUPRC and AUROC across 5-fold cross-validation on the GDPa1 dataset. The fine-tuned AbLang2-LoRA model (blue) is compared to a frozen AbLang2 embedding with Ridge regression classifier (orange). Both models use paired VH and VL input.

Table 5: Developability regression Spearman ρ (mean $\pm$ std, 5-fold cross-validation). Frozen AbLang2 + Ridge consistently outperforms LoRA fine-tuning across all targets, with the largest gaps on HIC and PR_CHO.

Target	AbLang2 + LoRA ρ	AbLang2 (frozen) + Ridge ρ	Ridge + p-IgGen CV ρ
HAC	0.536 $\pm$ 0.118	0.703 $\pm$ 0.077	0.621
AC-SINS_pH7.4	0.448 $\pm$ 0.156	0.523 $\pm$ 0.171	0.394
PR_Ova	0.366 $\pm$ 0.130	0.514 $\pm$ 0.120	0.549
Titer	0.294 $\pm$ 0.101	0.348 $\pm$ 0.136	0.239
SMAC	0.169 $\pm$ 0.252	0.325 $\pm$ 0.207	0.207
HIC	0.158 $\pm$ 0.109	0.447 $\pm$ 0.086	0.335
PR_CHO	0.117 $\pm$ 0.194	0.411 $\pm$ 0.093	0.496
T_{m2}	0.099 $\pm$ 0.174	0.134 $\pm$ 0.142	-0.084

OAS pretraining: The flow model was first pretrained on ~ 1 million HCDR3 sequences from the Observed Antibody Space (OAS) to learn the general CDR3 sequence manifold before fine-tuning on GDPa1; Figure 3 shows that GDPa1 training and test antibodies fall within the OAS convex hull in PLS space, confirming that the pretrained model has seen representative CDR3 diversity.

Binding Scorer Validation: Two antibodies without known human protein antigens (panobacumab, racotumomab) were excluded from all binding-related analyses. The binding scorer was trained in two stages on AbRank data. After Stage 1 pretraining, the held-out 10% validation Spearman ρ was 0.768. After Stage 2 fine-tuning, the validation Spearman ρ was 0.495. The Stage 1 metric reflects ranking ability across diverse antibody-antigen pairs; the Stage 2 metric reflects the harder task of distinguishing affinity differences among CDR3 variants binding the same antigen and framework, which is more directly relevant to our generation use case. The binding guidance term was included in the guided condition to steer CDR3 generation toward sequences with higher predicted affinity, but the scorer’s distributional mismatch with clinical mAbs means its scores are unreliable proxies for true binding affinity on this test set. We therefore do not report per-condition binding scores as a primary result.

Biophysical Property Evaluation of Generated Sequences: Table 6 compares four conditions across $n = 49$ held-out fold-1 antibodies: the parent (unmodified) sequence, the AbLang2-only baseline (unguided), the flow only guided generation, and the full guided generation (developability from the flow model and binding affinity from the prediction model). Four metrics are reported: oracle-predicted HIC and AC-SINS (the oracle’s scoring targets), CamSol solubility (a sliding-window hydrophilicity score applied to the full generated VH sequence), and sequence SAP score (a hydrophobic patch proxy applied to the full generated VH sequence).

Table 6: Biophysical evaluation of generative pipeline across $n = 49$ test antibodies (mean $\pm$ SE). Arrows indicate direction of improvement.

Metric	Parent	Baseline (AbLang2)	Flow-only	Guided (Flow + Binding)
Oracle HIC ($\downarrow$)	2.738 ± 0.048	2.535 ± 0.045	2.433 ± 0.046	2.411 ± 0.049
Oracle AC-SINS ($\downarrow$)	3.741 ± 1.215	-0.195 ± 1.165	-2.280 ± 1.149	-1.631 ± 1.234
CamSol ($\uparrow$)	0.150 ± 0.008	0.200 ± 0.011	0.217 ± 0.011	0.193 ± 0.011
SAP ($\downarrow$)	0.310 ± 0.002	0.300 ± 0.003	0.299 ± 0.002	0.305 ± 0.003

Table 7: Paired t -tests comparing generation conditions vs. parent ($n = 49$). Δ = mean change relative to parent. All tests are two-sided. * denotes $p < 0.001$.

Metric	Baseline vs. Parent		Flow Only vs. Parent		Flow + Binding vs. Parent	
	Δ	p	Δ	p	Δ	p
Oracle HIC	-0.203	< 0.001*	-0.305	< 0.001*	-0.326	< 0.001*
Oracle AC-SINS	-3.937	< 0.001*	-6.022	< 0.001*	-5.372	< 0.001*
CamSol	+0.051	< 0.001*	+0.067	< 0.001*	-0.044	< 0.001*
SAP	-0.011	< 0.001*	-0.012	< 0.001*	-0.006	< 0.0035

Paired t -tests reveal that all of the generation conditions achieve statistically significant improvements over the parent on all four metrics. The improvement on CamSol and SAP, which are not oracle training targets, warrants further investigation. However, since unguided AbLang2 generation alone produces comparable gains, these improvements likely reflect AbLang2’s language model prior favoring less hydrophobic CDR3 sequences rather than the flow model’s targeted optimization.

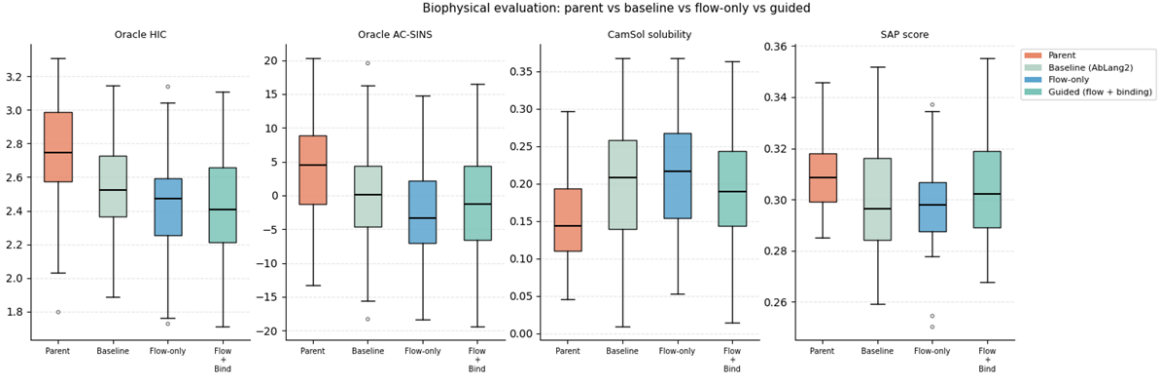


Figure 2: Biophysical evaluation across four conditions on the held-out test set ($n = 49$). Oracle HIC and AC-SINS ($\downarrow$ better); CamSol and SAP ($\uparrow$ and $\downarrow$ better, respectively).

5 Discussion

Our results reveal important insights about the use of antibody protein language models for developability prediction and generative sequence optimization. In addition, we note the fundamental challenges posed by small sample sizes and the high dimensionality of therapeutic antibody developability datasets.

5.1 Frozen Embeddings Outperform Fine-Tuning on Small Datasets

One of the main findings from Task A’s sequence-based scoring objective is that frozen AbLang2 embeddings paired with Ridge regression consistently outperform LoRA-fine-tuned AbLang2 in the majority of the developability regression targets. The frozen Ridge approach achieves a mean Spearman ρ improvement of +0.067 over the p-IgGen baselines reported by Arsiwala et al. [4], while LoRA fine-tuning yields a mean improvement of only -0.047. This gap is particularly striking for

developability targets with moderate signal: HIC ($\rho_{\text{Ridge}} = 0.447$ vs. $\rho_{\text{LoRA}} = 0.158$), PR_CHO (0.411 vs. 0.117), and SMAC (0.325 vs. 0.169).

This result is consistent with the bias-variance tradeoff in machine learning. With only $N = 197$ to 242 training samples per target and AbLang2’s 480-dimensional embedding space, LoRA fine-tuning introduces additional learnable parameters (the LoRA adapters plus a multi-layer regression head) that increase model capacity but also increase the chance for overfitting. Ridge regression, by contrast, applies strong L_2 regularization that shrinks coefficients toward zero, effectively limiting the model to the principal directions of covariance between embeddings and targets. The PCA dimensionality reduction applied before Ridge (capped at 48 components) further constrains the effective hypothesis class. This analysis suggests that working with small biophysical datasets should default to frozen PLM embeddings with simple linear models. Fine-tuning should be reserved for settings in which substantially more labeled data is available.

5.2 Target Coefficient of Variation Predicts Regression Difficulty

Error analysis identifies the coefficient of variation (CV) of each biophysical target as the strongest predictor of regression difficulty across both modeling approaches. Targets with $\text{CV} \leq 0.12$ (T_{m1} , T_{m2} , HIC) consistently yield Spearman $\rho < 0.16$ under LoRA fine-tuning, while high-CV targets such as AC-SINS_pH7.4 ($\text{CV} = 1.37$), PR_Ova ($\text{CV} = 1.15$), and PR_CHO ($\text{CV} = 0.91$) support substantially higher correlations. This relationship is not driven by sample size: HAC achieves the highest ρ (0.703 with Ridge) despite having the fewest samples ($N = 94$), while T_{m1} and T_{m2} have among the most samples ($N = 234$ and 193) but the worst performance.

One implication is that some developability properties are inherently more predictable from sequence than others. This may not necessarily be due to model limitations, but rather because the assay measurements themselves exhibit insufficient variance within the therapeutic antibody population. For example, SEC %Monomer clusters near 100% for nearly all antibodies in the data set. Given the high SEC %Monomer values across all antibodies, this could leave almost no signal for any model to learn from. This finding provides actionable guidance for future dataset design where training datasets are enriched with antibodies that span a wider range of biophysical property values. This would likely improve prediction performance more than increasing total sample count or model complexity.

5.3 Direct Approval Prediction from Sequence Alone Remains Challenging

Both frozen and LoRA fine-tuned AbLang2 models yield near-chance performance for binary approval classification ($\text{AUPRC} \leq 0.62$, $\text{AUROC} \leq 0.56$). This was substantially below the XGBoost upper bound trained on experimentally measured biophysical features (median $\text{AUPRC} = 0.665$). This gap may quantify the information loss incurred by replacing experimental assays with sequence-derived embeddings. While AbLang2 embeddings can encode meaningful biophysical signals for individual metrics (as demonstrated by the regression results), they do not capture enough of the complex landscape that determines developability and downstream clinical success.

Therapeutic antibody approval also depends on factors that go beyond biophysical properties. This includes target biology, patient selection, and manufacturing scalability (which are fundamentally absent from the raw amino acid sequence). Our isotype analysis further reveals that approval rates vary across IgG subtypes, suggesting that even within the biophysical domain, sequence-level representations may not capture isotype-specific manufacturing or pharmacokinetic considerations. Sequence-based approval prediction should therefore not be interpreted as a viable standalone tool. Instead, it may serve as one component of a multi-modal risk assessment framework that integrates clinical, biological, and biophysical features.

5.4 Generative Pipeline: AbLang2’s Prior Dominates Flow Steering

Our three-phase generative pipeline demonstrates statistically significant improvements in oracle-predicted HIC and AC-SINS scores, CamSol solubility, and SAP hydrophobicity relative to parent antibodies (paired t -test, $p < 0.05$). However, the unguided AbLang2 baseline, which samples CDR3 sequences from AbLang2’s masked language model head without any flow model steering, achieves comparable improvements to the guided condition. This suggests that AbLang2’s pretrained language model prior is the primary driver of improvement, not the conditional flow matching model.

Since AbLang2 was pretrained on millions of antibody sequences from the Observed Antibody Space, this implicitly encodes a prior over natural CDR3 sequences that avoids extreme hydrophobicity, aggregation-prone motifs, and other developability liabilities. When CDR3 positions are masked and resampled, the language model naturally proposes sequences that are more typical of the training distribution (and therefore more developable on average) than any individual therapeutic antibody’s CDR3, which may have been selected primarily for binding affinity rather than developability. The flow model’s contribution is attenuated because its steering signal operates in a 10-dimensional PLS space that partially captures the developability landscape.

5.5 Binding Affinity Guidance Contributes Negligible Steering

A central design goal of the guided condition was to steer HCDR3 generation toward sequences with improved developability and antigen-binding affinity, using the binding scorer trained on AbRank as an oracle during Phase 2 sampling. In practice, the binding guidance term contributed negligible differentiation between the guided and flow-only conditions. Several factors explain this failure.

First, only 47 of the 49 test antibodies had retrievable antigen sequences, and for several of these, the antigen sequence is a multi-meric complex or a conformational epitope that is not well captured by a single ESM2 chain embedding. The scorer’s cross-attention mechanism presupposes a single linear antigen chain, which limits its applicability to antibodies targeting structured or multi-meric antigens. Second, the per-residue ESM2 representations at Stage 2 provide richer antigen context than Stage 1 mean-pooling, but 8,000 examples are insufficient to reliably generalize to new antigen families. This means that the binding guidance term is noisy during Phase 2 sampling, and its contribution to the final joint score is dominated by the more reliable oracle-predicted developability signal.

5.6 OAS Pretraining and LoRA Oracle

Pretraining the flow model on ~ 1 million sequences from the Observed Antibody Space before fine-tuning on GDPa1 expands the learned CDR3 manifold (as visualized by the PLS space coverage analysis), showing that GDPa1 training points fall within the OAS convex hull. Since OAS sequences lack developability labels, pretraining uses zero-valued developability conditioning and allows the flow model to learn general CDR3 structure (but not developability-directed transport). The subsequent GDPa1 fine-tuning phase must learn the developability signal from only 197 augmented training points, which may be insufficient to override the pretrained velocity field.

The LoRA-fine-tuned oracle provides an alternative to the Ridge oracle for candidate reranking in Phase 3. While LoRA fine-tuning did not improve regression performance across the full set of 11 targets (Section 4), its use as a generative oracle is a different task. The oracle provides relative rankings among the $M = 20$ candidates generated for each parent antibody, rather than accurate absolute predictions.

5.7 Limitations

Some limitations should be considered when interpreting these results. First, all developability predictions and generative evaluations rely on computational oracles (Ridge or LoRA regression on AbLang2 embeddings) rather than experimental measurements. The oracle Spearman ρ values of 0.3–0.5 for HIC and AC-SINS indicate that oracle-predicted improvements may not fully translate to experimental improvements. Therefore, wet-lab validation of generated sequences is the necessary next step.

Another limitation stems from the size of the GDPa1 dataset (246 antibodies with paired developability data). This limits both the sequence scoring models (as evidenced by Ridge outperforming LoRA) and the generative flow model (which relies on data augmentation). In addition, our CDR3 extraction uses Aho numbering alignment, which may introduce errors for antibodies with unusual framework insertions or deletions.

Another limitation on the binding affinity prediction layer was the size of the AbRank dataset. The cross-attention binding scorer was trained on the AbRank dataset, which covers a broad but heterogeneous collection of antibody–antigen pairs that do not specifically represent the clinical mAbs in GDPa1. The Stage 2 fine-tuning set of about 8,000 multi-variant pairs is small, and the scorer’s Spearman ρ of ~ 0.31 on the held-out validation set indicates only moderate ranking ability.

As a result, the binding guidance signal during Phase 2 sampling is unreliable, and our evaluation cannot distinguish genuine binding improvements from scoring noise. Consequently, the contribution of binding guidance to the guided condition remains unquantified, and no claim is made that generated sequences preserve or improve experimental antigen-binding affinity. Resolving this limitation would require either a larger, therapeutics-focused binding dataset or experimental K_d measurements on the generated sequences themselves.

Finally, the generative pipeline optimizes only HCDR3 while keeping the framework and light chain fixed. While HCDR3 is the primary determinant of binding specificity and contributes substantially to developability, other CDRs and framework residues also influence biophysical properties. Extending the generative approach to jointly optimize multiple CDRs or full variable regions could yield additional improvements, but would require substantially more complex modeling.

5.8 Future Directions

Several directions could extend this work. One potential direction involves scaling the labeled training set through additional experimental measurements or using semi-supervised approaches that leverage the full OAS dataset to benefit both sequence scoring tasks and generative models (we ran out of compute, so expanding to the full dataset would be preferred). Another potential direction involves folding the generated antibody and using pretrained generative models that measure binding affinity between the folded antibody and the antigen, to ensure the model can also steer generation toward increased binding affinity. We could also replace the position-independent AbLang2 decoding with an autoregressive or iterative refinement scheme, which could improve the quality of generated CDR3 sequences by better capturing inter-residue dependencies.

6 Conclusion

We investigated whether antibody protein language models can support therapeutic antibody developability assessment in two complementary methods: (A) sequence-only prediction of biophysical properties and approval, and (B) guided HCDR3 sequence generation using conditioned flow matching models.

For Task A, frozen AbLang2 embeddings with Ridge regression consistently outperformed LoRA fine-tuning of AbLang2 across 11 biophysical targets under hierarchical-cluster cross-validation (mean $\Delta\rho = +0.067$ over the p-IgGen baseline), with the strongest signals on HAC ($\rho = 0.703$), AC-SINS_pH7.4 ($\rho = 0.523$), and PR_Ova ($\rho = 0.514$). Sequence-only approval classification yielded near-chance performance (AUPRC ≤ 0.62 , AUROC ≤ 0.56) compared to an experimental-feature upper bound of 0.665, quantifying the information gap between sequence and the multifactorial determinants of clinical success.

For Task B, our three-phase generative pipeline produced HCDR3 variants with statistically significant improvements in oracle-predicted HIC and AC-SINS, CamSol solubility, and SAP score relative to parent antibodies. However, comparisons against an unguided AbLang2 sampling baseline reveal that AbLang2’s pretrained prior, not flow-based steering, drives most of the observed improvement. This indicates that current flow-matching architectures provide limited benefit over the implicit developability prior encoded in OAS-pretrained language models at these sample sizes.

The most immediate next step is experimental validation of generated sequences, as oracle Spearman correlations of 0.3–0.5 mean computational gains may not fully transfer to wet-lab measurements. Beyond this, three directions are promising: scaling labeled developability datasets, incorporating antigen-specific conditioning into the flow model, and extending generation beyond HCDR3 to jointly optimize multiple CDR regions.

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Appendix

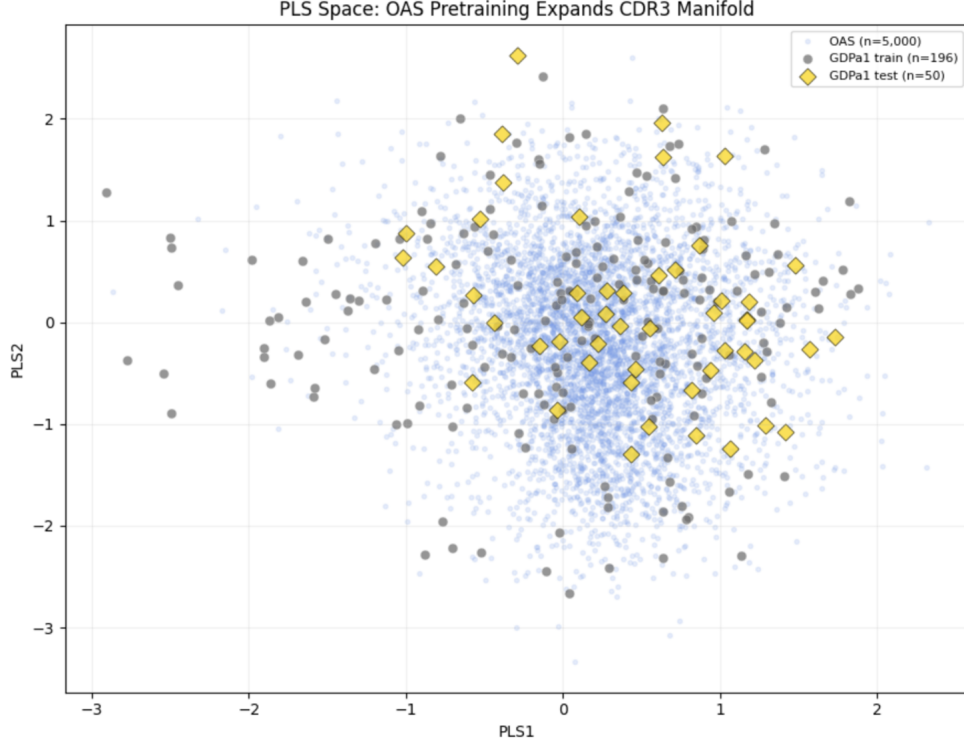


Figure 3: PLS space visualization showing that OAS pretraining expands the learned CDR3 manifold. A random sample of 5,000 OAS sequences (blue) broadly covers the PLS space, while GDPa1 training antibodies (gray, $n = 196$) and test antibodies (yellow, $n = 50$) fall within the OAS convex hull.

A ODE Update Equation

At each Euler step n , the PLS position is updated as:

$$x_{n+1} = x_n + \Delta t \left(v_\theta(t_n, x_n) + \lambda \cdot \frac{\nabla_x s_{\text{bind}}(f^{-1}(x_n))}{\|\nabla_x s_{\text{bind}}(f^{-1}(x_n))\|} \right), \quad (2)$$

where $x_n \in \mathbb{R}^{10}$ is the antibody’s position in PLS-reduced CDR3 embedding space at step n (initialized from $x_0 \sim \mathcal{N}(0, I)$), v_θ is the trained flow model velocity pointing toward the high-developability region of PLS space, $f^{-1} : \mathbb{R}^{10} \rightarrow \mathbb{R}^{480}$ is the PLS inverse projection back to AbLang2 embedding space, s_{bind} is the cross-attention binding scorer predicting $-\log K_d$, $\Delta t = 1/100$, and $t_n = n \cdot \Delta t$. The gradient is unit-normalized so λ controls a fixed step size in the binding ascent direction.

For *guided* condition $\lambda=0.5$ is used and for the *flow-only* condition $\lambda=0$ is used.